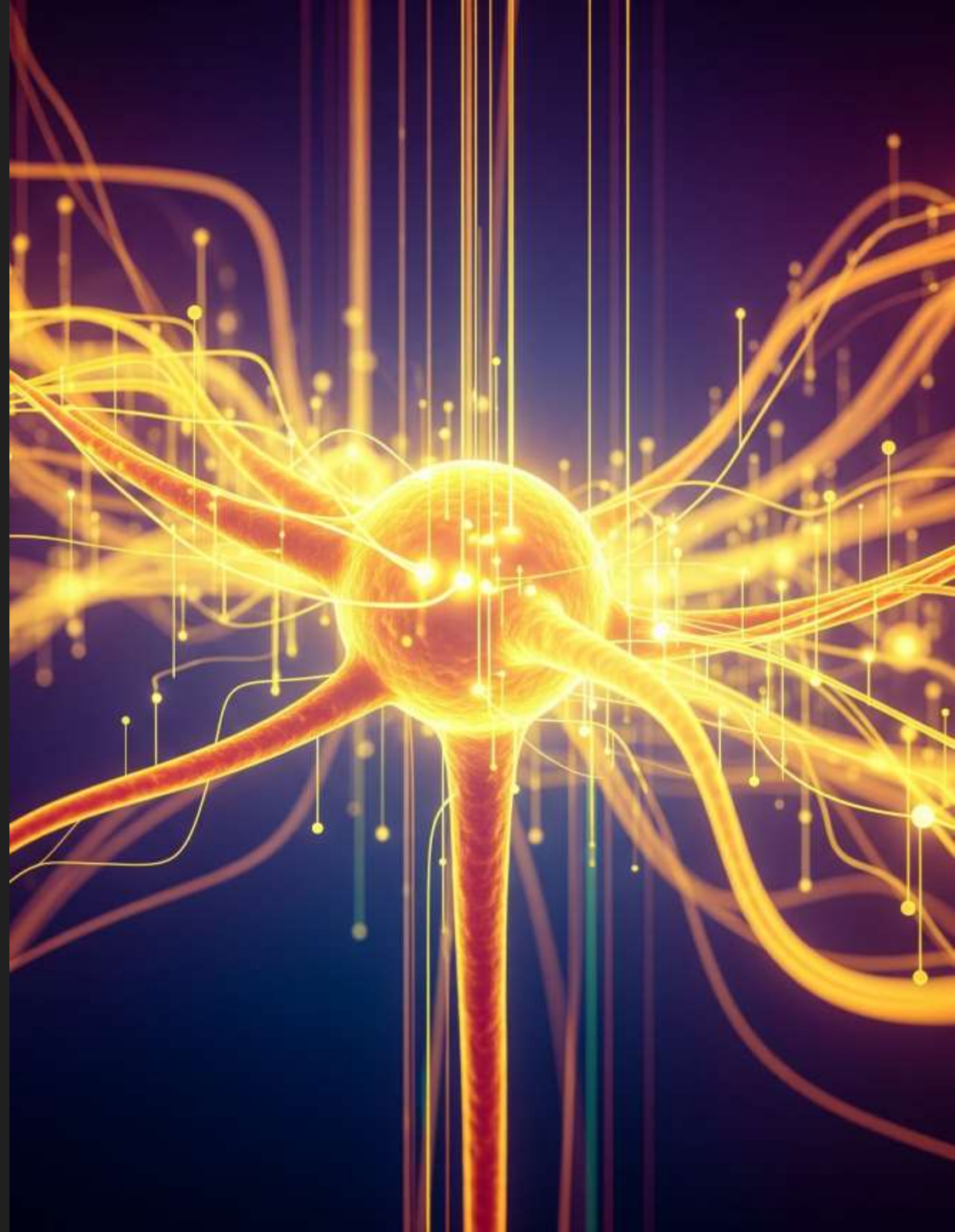
Citrus Classification with Perceptron

This presentation explores using a Perceptron model to classify citrus fruits. We will cover the problem, data, technique, and results.

Student : Hamedallah Anwer Hamedallah Issa

Student ID : 320220603007



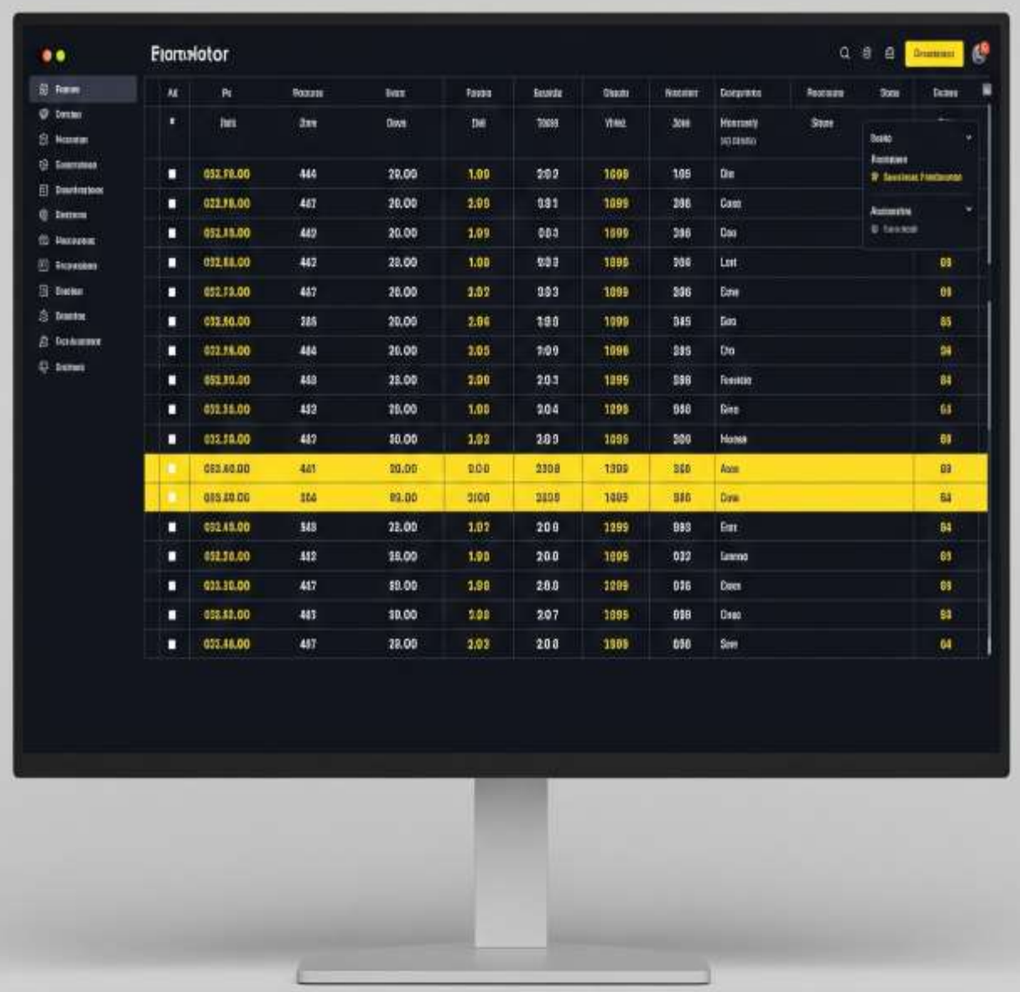
Problem Statement and Motivation

The Challenge

Accurately classify citrus fruits based on physical attributes. Distinguishing between oranges and grapefruits can be difficult.

Our Goal

Develop a simple, effective model for automated fruit sorting. This improves efficiency in agricultural processes.



Dataset Description

Citrus Dataset

Contains features like diameter, weight, and RGB color values. Each entry is labeled as either 'orange' or 'grapefruit'.

Data Generation

The dataset was generated from average orange and grapefruit characteristics. This creates a diverse range of values for training.

Perceptron Technique Explained

Simple Neuron

The Perceptron is a fundamental building block of neural networks. It makes binary decisions.

Weighted Sum

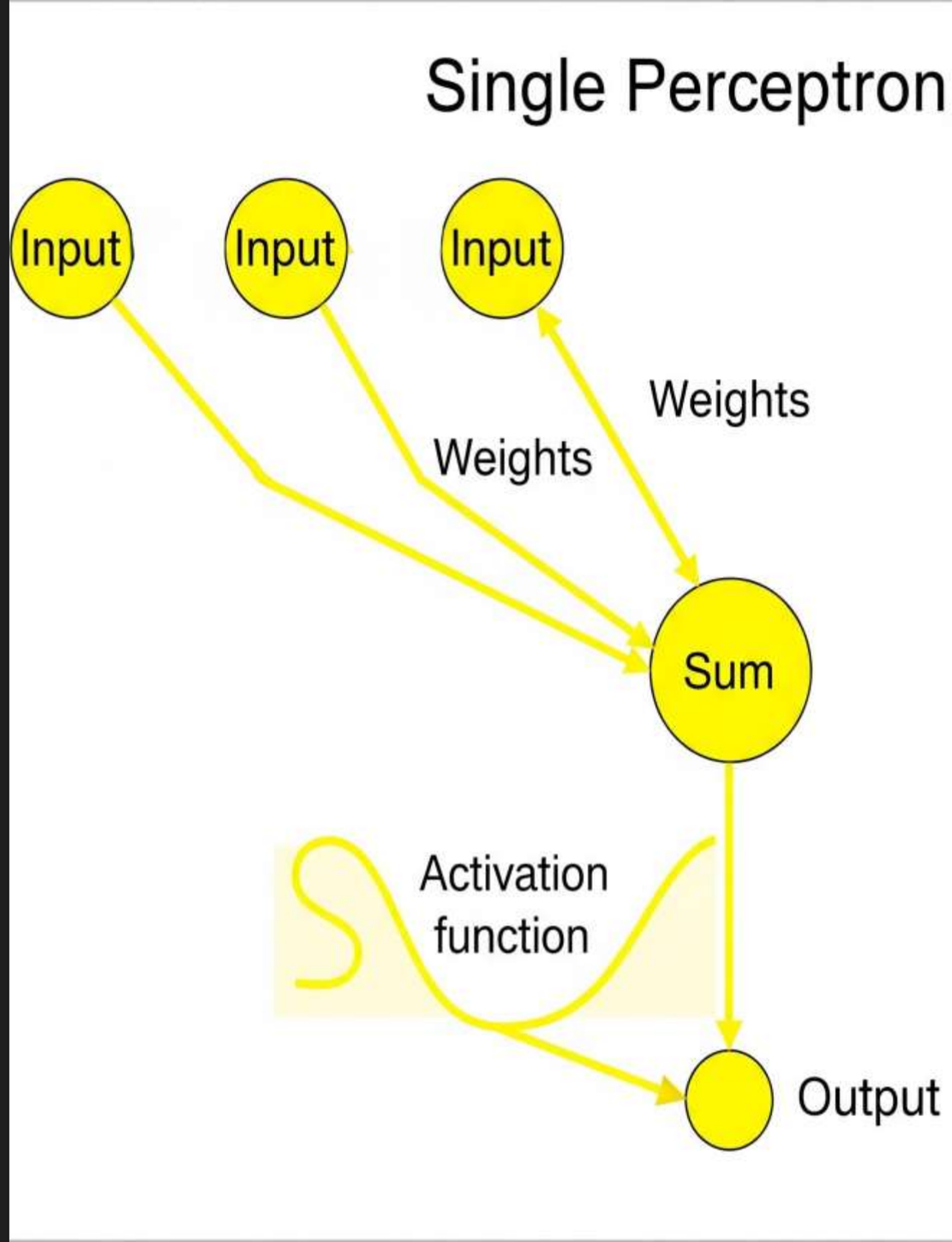
It calculates a weighted sum of inputs. A bias is added to this sum.

Activation Function

A step function determines the output. It outputs 1 or 0 based on the sum.

Learning Rule

Weights and bias are adjusted based on prediction errors. This process iterates to improve accuracy.



Data Preprocessing Steps

1

Load Data

The "citrus.csv" file is loaded into a Pandas DataFrame.

2

Encode Labels

Fruit names ('orange', 'grapefruit') are converted to numerical values (0, 1) using LabelEncoder.

3

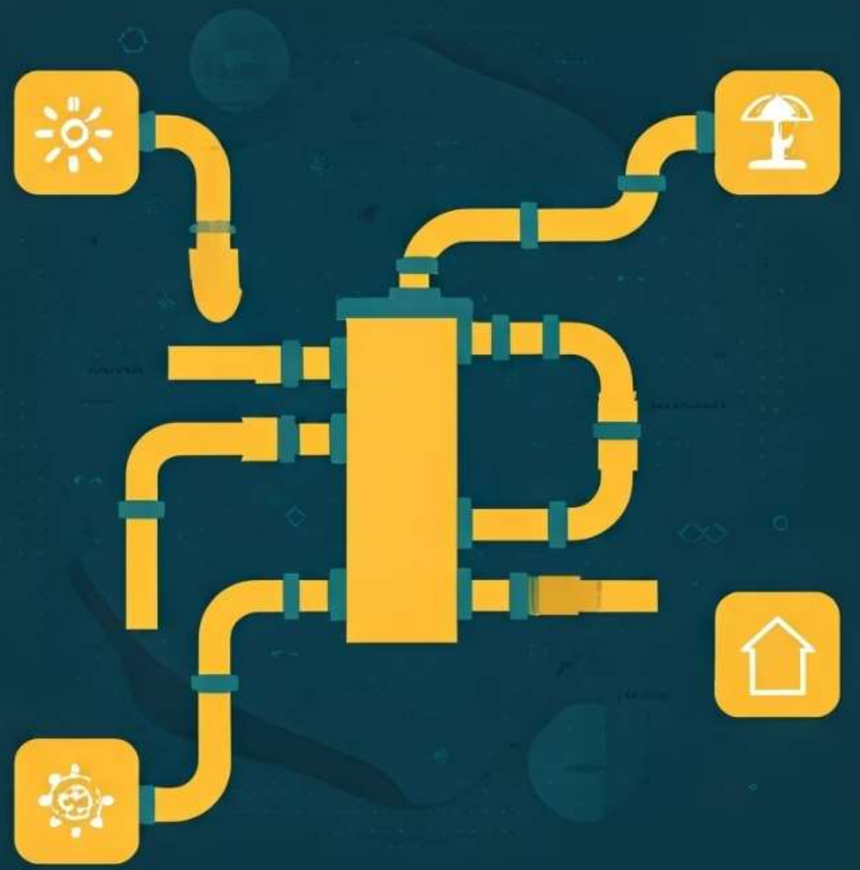
Split Data

The dataset is divided into training and testing sets. This ensures unbiased model evaluation.

4

Scale Features

Numerical features are scaled using StandardScaler. This prevents features with larger values from dominating.





Model Training and Evaluation

1 Model Initialization

A Perceptron instance is created with a learning rate of 0.01 and 1000 iterations.

2 Training Process

The model is trained on the scaled training data. It adjusts weights and bias iteratively.

3 Prediction

Predictions are made on the unseen, scaled test data. This assesses generalization ability.

4 Accuracy Calculation

Accuracy is computed by comparing predictions to true labels. The result is printed to evaluate performance.

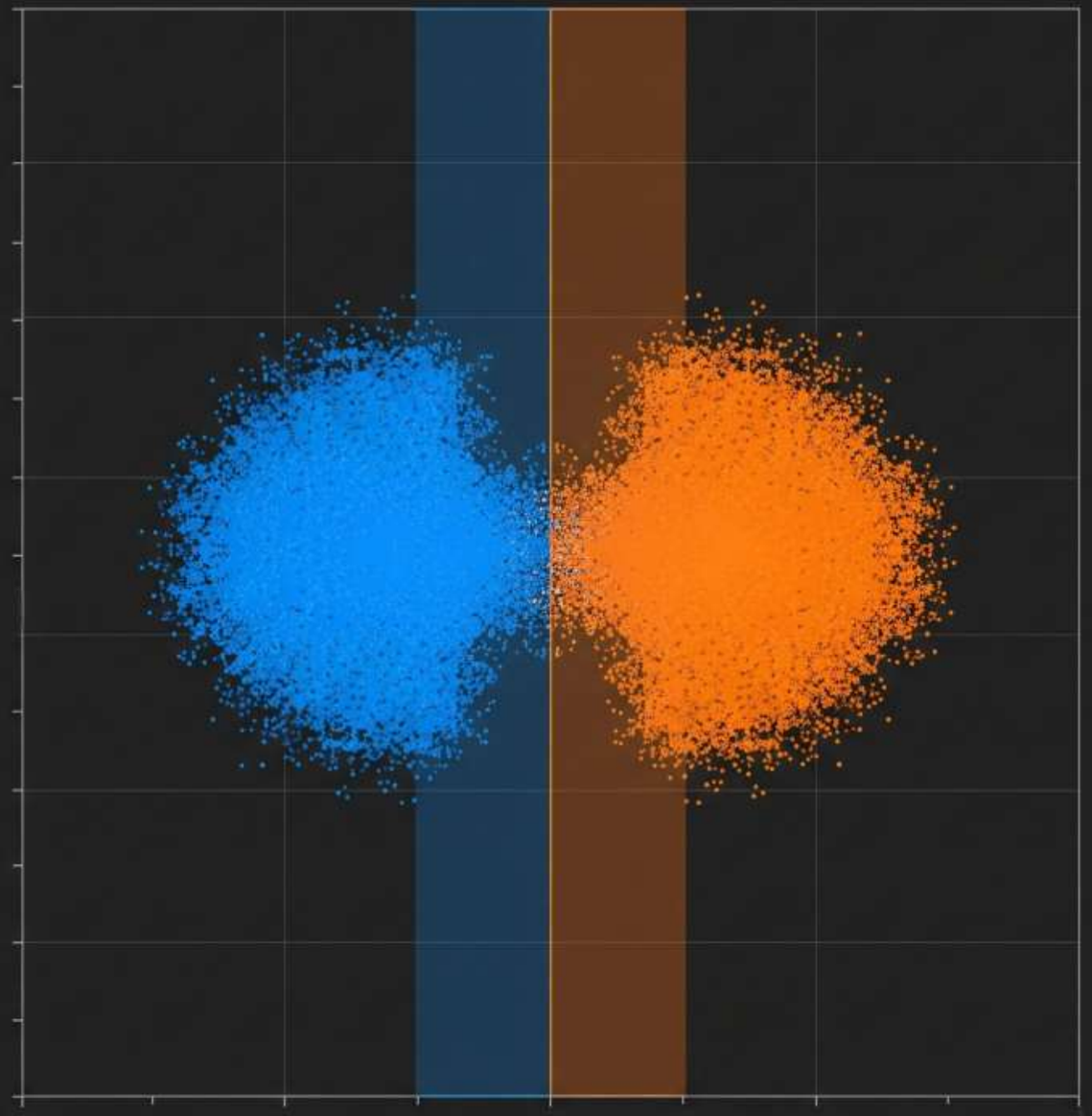
Results and Evaluation

93.3%

Accuracy

The Perceptron achieved high accuracy on the test set. This indicates strong classification performance.

The model successfully learned to distinguish between oranges and grapefruits. This demonstrates the effectiveness of the Perceptron for binary classification tasks.



Conclusion and Next Steps

Summary

We successfully built and evaluated a Perceptron model. It accurately classifies citrus fruits.

Future Work

Explore more complex models like Multi-Layer Perceptrons. Consider additional features for enhanced accuracy.

Applications

This model can be integrated into automated sorting systems. It has potential for various agricultural uses.

